Table 7.1 System Testing Results

SL No	Description	Input	Expected Output	Actual Output	Status
1	Line sensor test	Black path on white surface	Robot follows path accurately	Robot follows smoothly	Pass
2	RFID scanning	NFC-tagged waste near reader	Tag detected, ID displayed	ID on serial monitor	Pass
3	Twilio SMS alert	NFC violation detected	SMS sent to violator	SMS in 8 seconds	Pass
4	GPS verification	System powered outdoors	Accurate coordinates displayed	Correct SMS coordinates	Pass
5	Conveyor operation	NFC tag detected	Belt moves for 3 seconds	Belt activated as expected	Pass
6	IR sensor failure	One sensor disconnected	System halts, gives alert	Alert/motion stopped	Pass
7	Motor direction test	Forward command	Robot moves forward	Moves as expected	Pass
8	ESP32 communication	Wi-Fi connected	Data sent to Twilio API	Twilio confirms delivery	Pass
9	System integration	All modules connected	Synchronized operation	Stable integration	Pass